Teaching XSLT for Digital Arts and Humanities in the Age of AI

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Our word processors and software tools now regularly deliver prefabricated constructions that subtly guide the writing process. Word processing software and chat interfaces on our phones deliver automated grammar and syntax checking, suggesting to us the best-fit next words and phrases gently enforcing normalized expectations drawn from a vector database. This all-too-familiar domain of our technological dependencies is part of what David Berry has described as manipulative "infrasomatization" that "saturates everyday life," priced and delivered to maintain a dependent consumer base (Berry par. 15). Berry suggests that providing a satisfactory basis for "explanation" of invasive algorithms in the cause of reducing human anxiety may always be insufficiently simplistic or abstract. In the practice of computational work in humanities, establishing trustworthy and reliable grounds of scholarly authority requires that we make critical decisions about the basis for establishing a reliable digital editions and responsible practice with preparing data that can be replicated and critiqued. With the dominance of large language models, the grounds of verifiable knowledge are increasingly unclear in public media, and nevertheless increasingly vital to for us to discover, especially when educating students in digital arts and humanities.

This paper argues that the practice of markup languages is a powerful educational tool in digital humanities coursework because it foregrounds the modeling of data from documents and the transference of media formats in a manner that can be documented. In learning XSLT, specifically, students gain authority over their means of production, an authority that is lacking in the word processing environment that dominates the production of essays in much of humanities coursework. Any hands-on engagement with media production (from typesetting equipment at the Skeuomorph lab to the engagement in zine construction) involves students in a careful transfer of data from one structure to another. But engagement with XML markup and the tooling designed for it, particularly with XSLT, eXtensible Stylesheet Language Transformations, affords students a critical opportunity to step away from constraining boxes to design systematically and aesthetically on their own terms, and also a means to transfer and mediate data in a design that they establish. XSLT becomes a design tool for students to explore how to create and manage meaningful containers of information and the basis for interactivity with their projects. In a digital media program, XSLT becomes a mode of graphic and layout production, a tool for data extraction amenable to creative play and multiple output possibilities from a single source. We more frequently *teach* markup and XSLT than *theorize or reflect* about the power of these tools for humanities practice, but one teaching resource from 2022 gives voice to the excitement of freedom to work with a powerful tool on one's owns terms as "Ninja'ing" when writing XSLT for the Web and in LaTeX (https://latex-ninja.com/tag/xslt/).

While student markup is not without flaws, the best projects take on a life of their own as a basis for citizen science and public-facing scholarship that can emerge organically from student's own interests. To demonstrate this, we propose to share student projects from our lab (curated at newtfire.org) that engage in document and metadata curation working with GitHub repositories and sharing data from historic comic books to movie scripts and game scripts. We will concentrate on the affordances of XSLT in these projects to help elaborate how students take control of re-mediation and data transfer in the design process. Students in the "Digit" program at Penn State Behrend encounter the XML stack as building blocks for the construction of digital archives and synthesize fundamental skills of creative design, systematic analysis, and descriptive explanation grounded in experience of working with a community of practice. In the context of the skills they gain with structured markup, students continue to learn applications of Python and regular expressions that treat the text as strings and streams (including markup itself as a pattern). They gain experience in core classes in our program with XML, TEI, HTML, CSS, JavaScript, SVG, Relax NG schema languages, XSLT, and XQuery, while taking coursework as they choose to specialize in graphic design, data visualization, or videography, digital audio, and game design. Each course in this program involves design and production of media, and in that context, courses in Text Encoding and Text Analysis are part of a makerspace environment that offers grounding data and programming as well as design.

In this context in a "makerspace" major for digital creatives, XSLT looks different from something one just chooses to adapt or copy. It beco-

mes a basis for a design paradigm for analyzing sources and outputting new designs that do not have to follow a controlled content management system. Students gain graphic design skills in writing XSLT together with CSS stylesheets and (later in the program) with JavaScript as a means to plan and develop interfaces on their own terms. As students are learning to become producers in the context of their own projects, perhaps they have a chance to experience what Malazita describes as a goal for DH laboratories: Ideally, we are creating an "epistemic culture" within the university, a space for students to become humanities "knowledge workers."

Recently I asked Google Gemini what it thought of teaching XSLT to undergraduates. It answered that it can improve understanding of XML, data transformation, and translatable programming concepts, but that it may be too specialized and difficult to learn. When I replied, "What if the context is not computer science but digital humanities?" Gemini was more positive, citing the "flexibility and power" of this programming language and its wide usage in DH, though again it repeated the "steep learning curve" caveat. XSLT is certainly a technology better known to humanities scholars who work with digital scholarly editions, but while shifting the context mattered in addressing the AI, we should note that even in discussions of the programming available for text processing, humanities contexts must be specified or they disappear, apparently irrelevant. Perhaps that emphasizes what's at stake for authority over technology in the humanities. A digital humanities coursework that sets markup practice at the center of students' skillset may serve as a strong foundation for us to explore an alternative basis for understanding than currently available in AI-driven mainstream technologies.

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